LAB 1

DevOps Foundations & Continuous Integration

Task 5 - Set Up Version Control Using Git

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Objective

Initialize a Git repository, stage and commit changes, create and work with a branch, merge the branch into main, configure a GitHub remote, push the repository to GitHub, and verify the final repository status.

Project: ~/lab1-devops **Remote:** GitHub lab1-devops-version-control

Evidence sequence

The screenshots are arranged chronologically so each image directly supports the corresponding step of the required Git version-control workflow.

1. Repository location and initial project file

The working directory is **~/lab1-devops**. The **README.md** file is present, establishing the project before Git initialization.

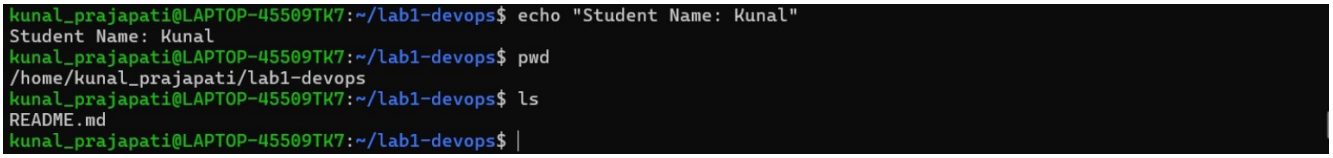

Figure 1. 1. Repository location and initial project file.

2. Initialize the Git repository

The command **git init** initializes the local Git repository. **git status** confirms the repository is on the initial **master** branch with README.md untracked.

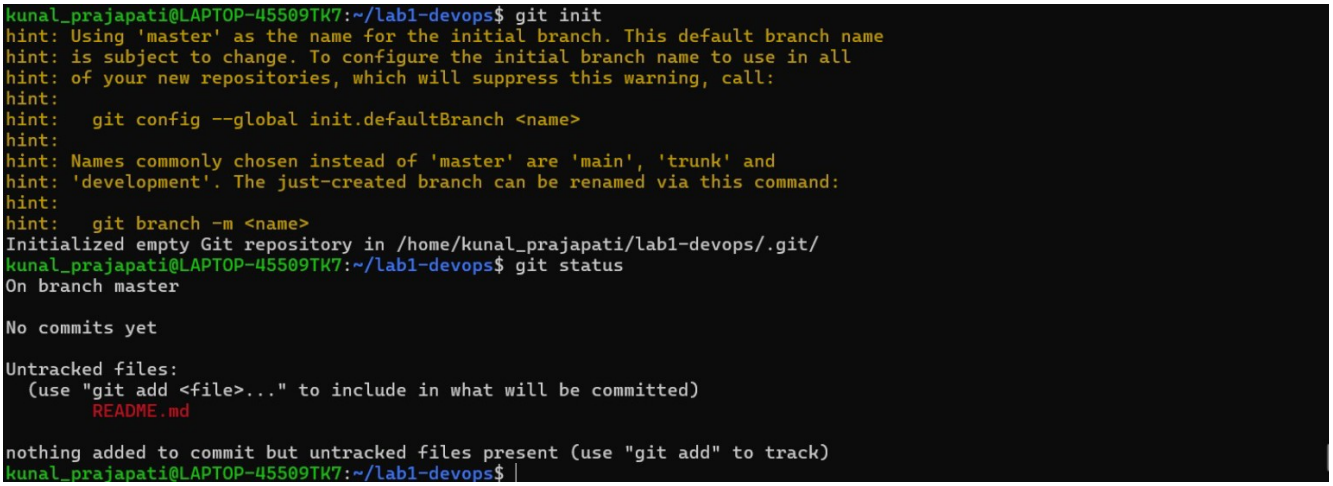

Figure 2. 2. Initialize the Git repository.

3. Stage files for commit

The command **git add README.md** stages the project file. The status output shows README.md under **Changes to be committed**.

```
kunal_prajapati@LAPTOP-45509TK7:~/lab1-devops$ git add README.md
kunal_prajapati@LAPTOP-45509TK7:~/lab1-devops$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md

kunal_prajapati@LAPTOP-45509TK7:~/lab1-devops$
```

Figure 3. 3. Stage files for commit.

4. Create the initial commit

The command **git commit -m "Initial project setup"** creates the first commit. The Git log confirms commit **ee06e58**.

```
kunal_prajapati@LAPTOP-45509TK7:~/lab1-devops$ git commit -m "Initial project setup"
[master (root-commit) ee06e58] Initial project setup
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
kunal_prajapati@LAPTOP-45509TK7:~/lab1-devops$ git log --oneline
ee06e58 (HEAD -> master) Initial project setup
kunal_prajapati@LAPTOP-45509TK7:~/lab1-devops$
```

Figure 4. 4. Create the initial commit.

5. Create a feature branch and commit changes

The **feature-notes** branch is created and a version-control note is committed. The log shows both the feature branch commit and the initial main/master history.

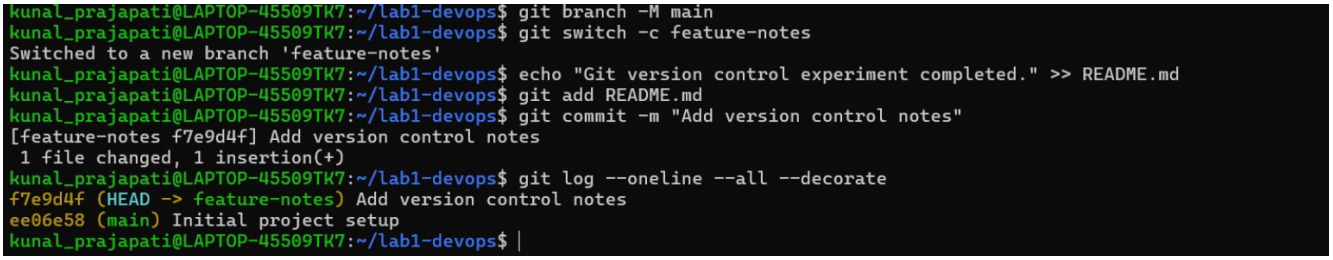

Figure 5. 5. Create a feature branch and commit changes.

6. Merge the feature branch into main

The terminal shows **git switch main** followed by **git merge feature-notes**. The merge completes successfully and the graph shows **HEAD -> main** at the feature commit.

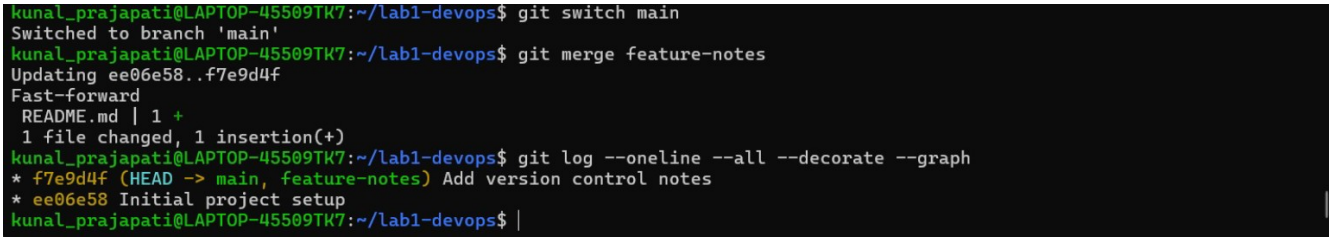

Figure 6. 6. Merge the feature branch into main.

7. Configure the GitHub remote

The GitHub repository is configured as the **origin** remote. **git remote -v** verifies both fetch and push URLs.

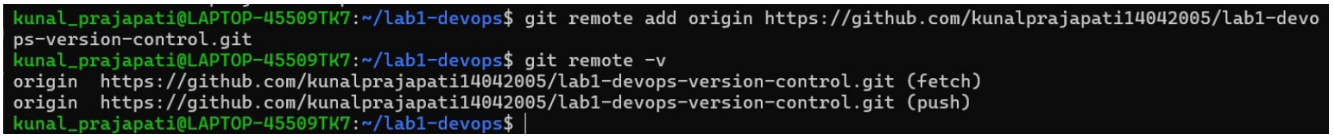

Figure 7. 7. Configure the GitHub remote.

8. Push the repository to GitHub

The **git push -u origin main** command successfully pushes the local main branch to GitHub and sets **origin/main** as the upstream branch.

Figure 8. 8. Push the repository to GitHub.

9. Final repository status

The final **git status** confirms the local main branch is up to date with **origin/main** and the working tree is clean.

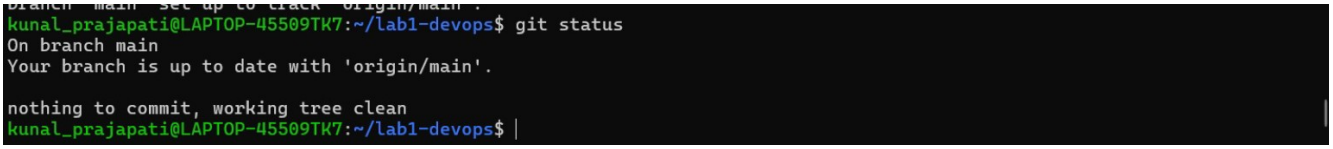

Figure 9. 9. Final repository status.

Conclusion and Requirement Mapping

Task 5 requirement: initialize a repository, stage and commit changes, create branches, merge branches, and push the repository to GitHub.

Required activity	Evidence
Initialize repository	Figure 2 - git init and git status
Stage changes	Figure 3 - git add and staged status
Commit changes	Figure 4 - initial commit and log
Create branch	Figure 5 - feature-notes branch
Merge branch	Figure 6 - switch main and merge feature-notes
Configure GitHub remote	Figure 7 - git remote -v
Push to GitHub	Figure 8 - successful git push
Verify final state	Figure 9 - clean, up-to-date git status

Conclusion

Task 5 was completed successfully. The evidence demonstrates the complete local Git workflow from repository initialization through staging, commits, branching, merging, remote configuration, GitHub push, and final clean-status verification.